Requirements & Solution Planning

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Session Topics

- ✔ Convert business needs into AI requirements
- ✔ Define inputs, outputs, users, and workflow
- ✔ Identify data sources and data-quality requirements
- ✔ Define human intervention and approval points
- ✔ Develop functional and non-functional requirements
- ✔ Establish measurable success criteria
- ✔ Collaborate with AI, IT, and vendor teams
- ✔ Prepare an **AI Project Requirement Brief**

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Converting Business Needs into AI Requirements

Business Need vs AI Requirement

Business Need

Describes **what the organisation wants to achieve**.

Example:

"We want to reduce customer support response time."

AI Requirement

Describes **what the AI solution needs to do** to achieve that outcome.

Example:

"The AI system should classify incoming customer queries, identify their priority, and recommend an appropriate response."

Key Difference

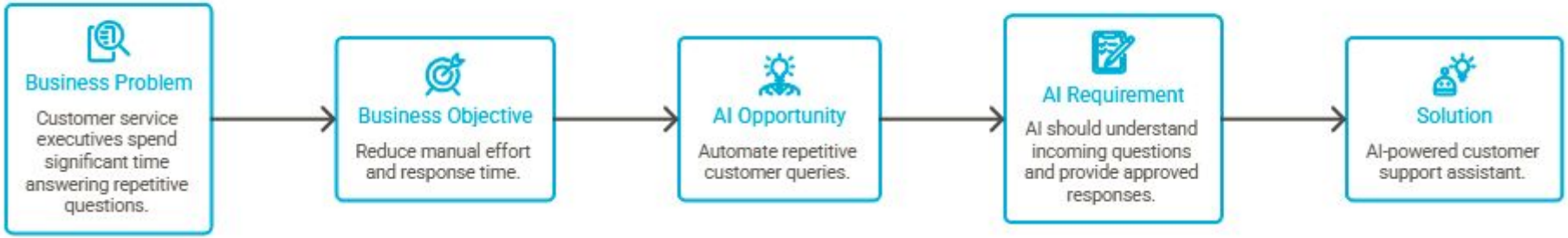
Business Need	Reduce response time	Improve customer experience	Reduce manual effort	Improve decision-making
AI Requirement	Automatically classify queries	Generate response recommendations	Automate repetitive questions	Provide AI-based recommendations

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From Business Problem to AI Solution



Important Questions

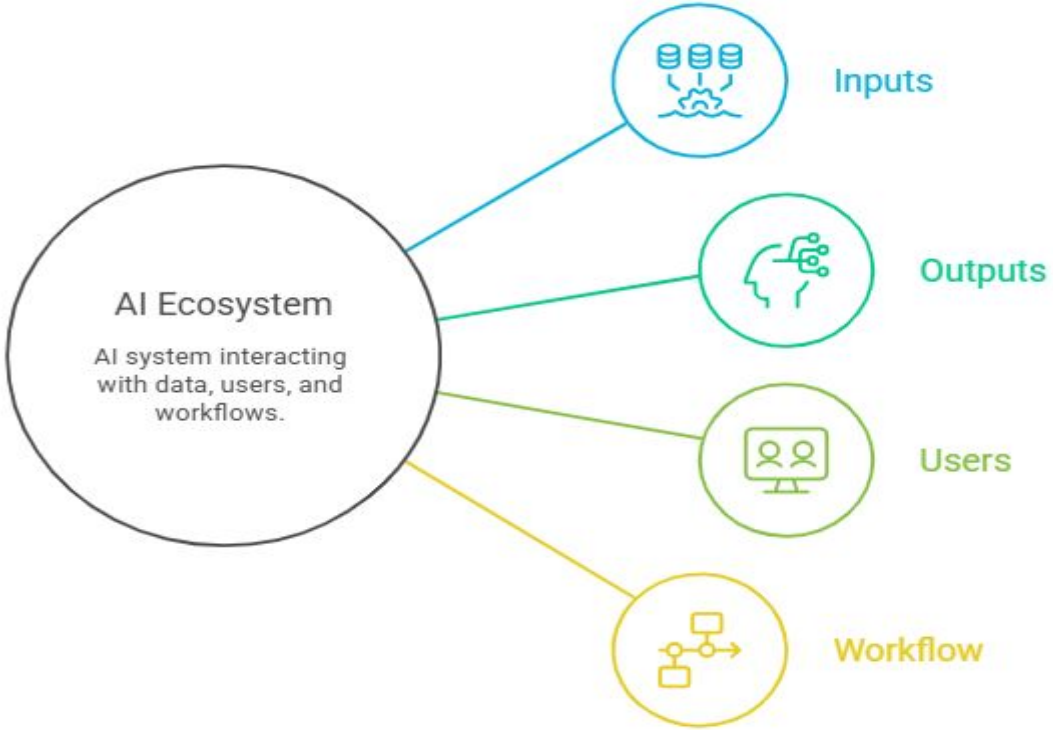
- ✔ What problem are we solving?
- ✔ Who experiences the problem?
- ✔ How frequently does it occur?
- ✔ What is the current process?
- ✔ Where can AI add value?
- ✔ What should AI actually do?

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Inputs, Outputs, Users & Workflow



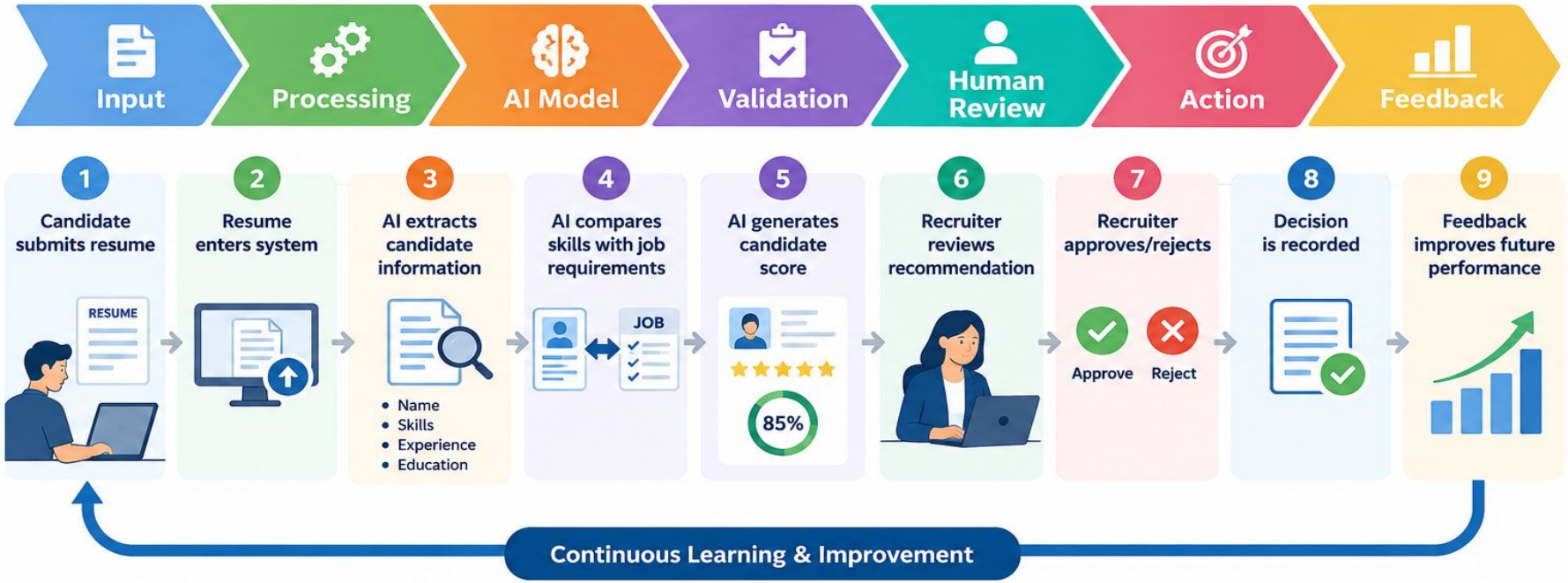
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Map the AI Workflow

Example: AI Recruitment Screening



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Data Sources & Data Quality : Identify Data Sources

Common Internal Data Sources

- **CRM** — Customer and sales data
- **ERP** — Finance, operations, supply chain
- **HR systems** — Employee and talent data
- **Customer databases** — Customer profiles and interactions
- **Transaction systems** — Orders, payments, invoices
- **Documents** — Reports, policies, contracts
- **Emails** — Internal and customer communications
- **Call recordings** — Customer support calls
- **Historical reports** — Business performance data

Common External Data Sources

- **Public datasets** — Government, research, open data
- **Market data** — Industry trends, competitor data
- **APIs** — Real-time data from external services
- **Third-party databases** — Specialised commercial data
- **Vendor-provided data** — Partner or supplier data



Data Source Assessment

Ask:

- Where is the data?
- Who owns it?
- How frequently is it updated?
- Can the AI system access it?
- Is permission required?
- Is the historical data sufficient?

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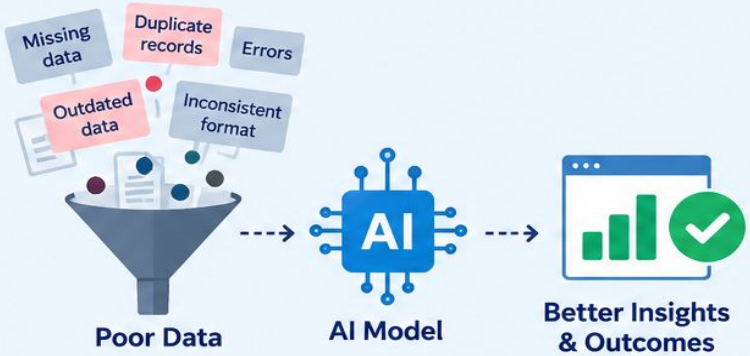
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DATA QUALITY REQUIREMENTS

Why Data Quality Matters

Poor Data → Poor AI Output

AI performance depends heavily on the quality, relevance, and consistency of its data.



Important Data Quality Dimensions

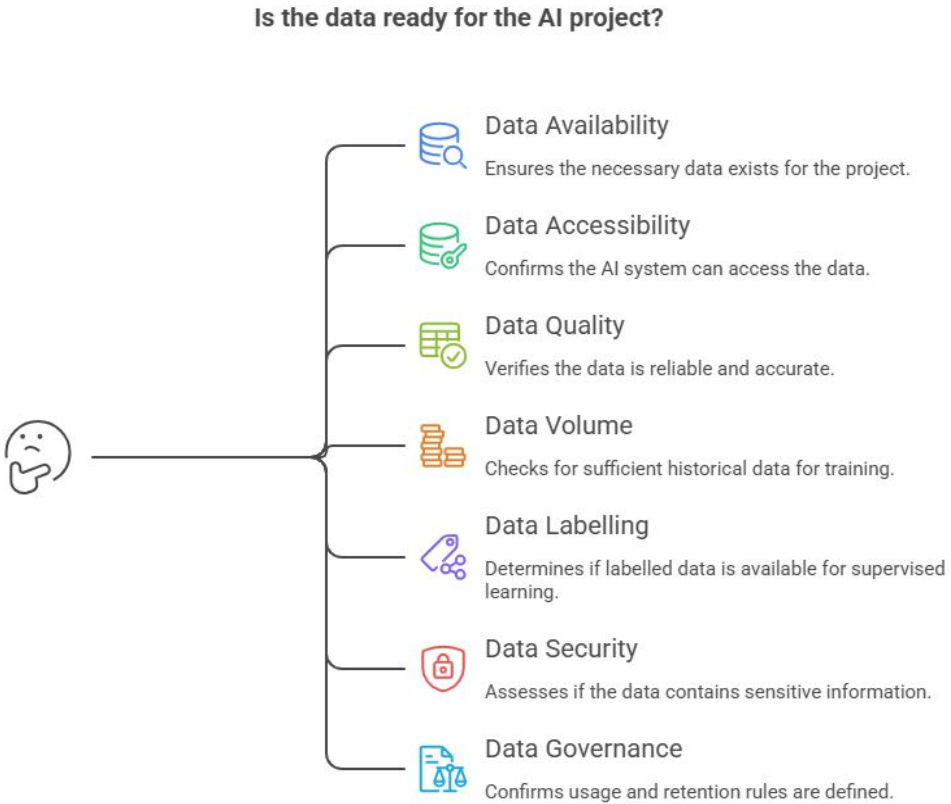
Dimension	Key Question	Icon
Accuracy	Is the data correct?	
Completeness	Are important fields missing?	
Consistency	Is data formatted consistently?	
Timeliness	Is the data current?	
Relevance	Does it support the use case?	
Uniqueness	Are duplicate records present?	
Validity	Does the data follow expected rules?	

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Data Readiness Assessment



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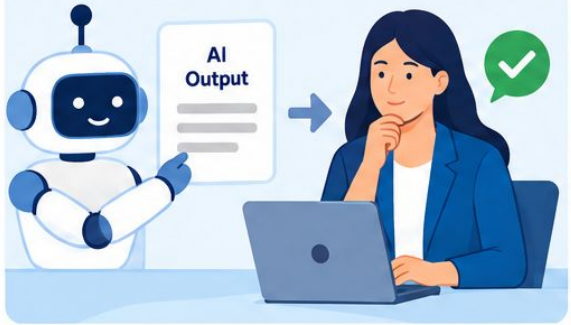
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Human Intervention & Approval

What is Human-in-the-Loop?

A human reviews, validates, approves, or overrides AI-generated results before important actions are taken.



Example Workflow

AI (Analysis)
AI processes data and provides results

→ **Recommendation**
AI generates a recommendation or prediction

→ **Human Review**
A human reviews and validates the result

→ **Approval**
Human approves (or overrides) the result

→ **Action**
Approved result is used to take a business action

Why Human Intervention?

- AI can make mistakes
- Some decisions have high business impact
- Regulatory requirements may require human oversight
- Customers may need human support
- Exceptions require human judgment

Examples

- Loan approval
- Recruitment decisions
- Medical recommendations
- Legal document review
- Financial transactions

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Define Approval Points

Identify Where Humans Are Required

For every AI workflow, ask:

"Can the AI make this decision independently?"

If not:

AI Recommendation → Human Approval

Three Levels of Human Control

Level	AI Role	Human Role
Level 1 – Human-in-the-Loop	AI recommends	Human approves
Level 2 – Human-on-the-Loop	AI acts automatically	Human monitors performance
Level 3 – Human-out-of-the-Loop	AI operates autonomously	None

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Functional & Non-Functional Requirements

Functional Requirements
What the AI System Must Do

Functional requirements describe the specific capabilities and actions of the solution.

Example: AI Customer Support Assistant

1. System should classify customer queries
2. System should identify query priority
3. System should retrieve relevant knowledge
4. System should generate response suggestions
5. User should be able to edit AI responses
6. System should record user approval
7. System should escalate complex queries

Non-Functional Requirements
How the System Should Perform

Non-functional requirements define system quality, performance, security, and operational expectations.

Key Categories

- Performance
- Security
- Scalability
- Reliability
- Availability
- Usaibility
- Privacy
- Maintainability
- Compliance

Functional vs Non-Functional
Example Comparison

Functional Requirement (What)	Non-Functional Requirement (How Well)
Classify customer queries	Response within 3 seconds
Generate recommendations	99.5% system availability
Summarise documents	Secure access
Create response drafts	Support 10,000 users
Escalate complex cases	Maintain audit logs

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Measurable Success Criteria:

Why Define Success Criteria?

Without measurable criteria, it is difficult to determine whether the AI project actually created value.

Avoid Vague Goals

❌ "Improve customer service."

✅ **Instead:**

"Reduce average customer response time by 30%."

Success Metrics

Business Metrics

- Revenue increase
- Cost reduction
- Productivity
- Conversion rate

Operational Metrics

- Processing time
- Automation rate
- Error rate
- Throughput

AI Metrics

- Accuracy
- Precision
- Recall
- Response quality
- Hallucination/ error rate

User Metrics

- Adoption
- Satisfaction
- User acceptance

KPI Example : AI Customer Support Assistant

Baseline (Current State)	Target (Desired State)
Average response time **10 minutes**	Response time **< 5 minutes**
Manual handling **100%**	AI-assisted interactions **70%**
Customer satisfaction **75%**	Customer satisfaction **> 85%**

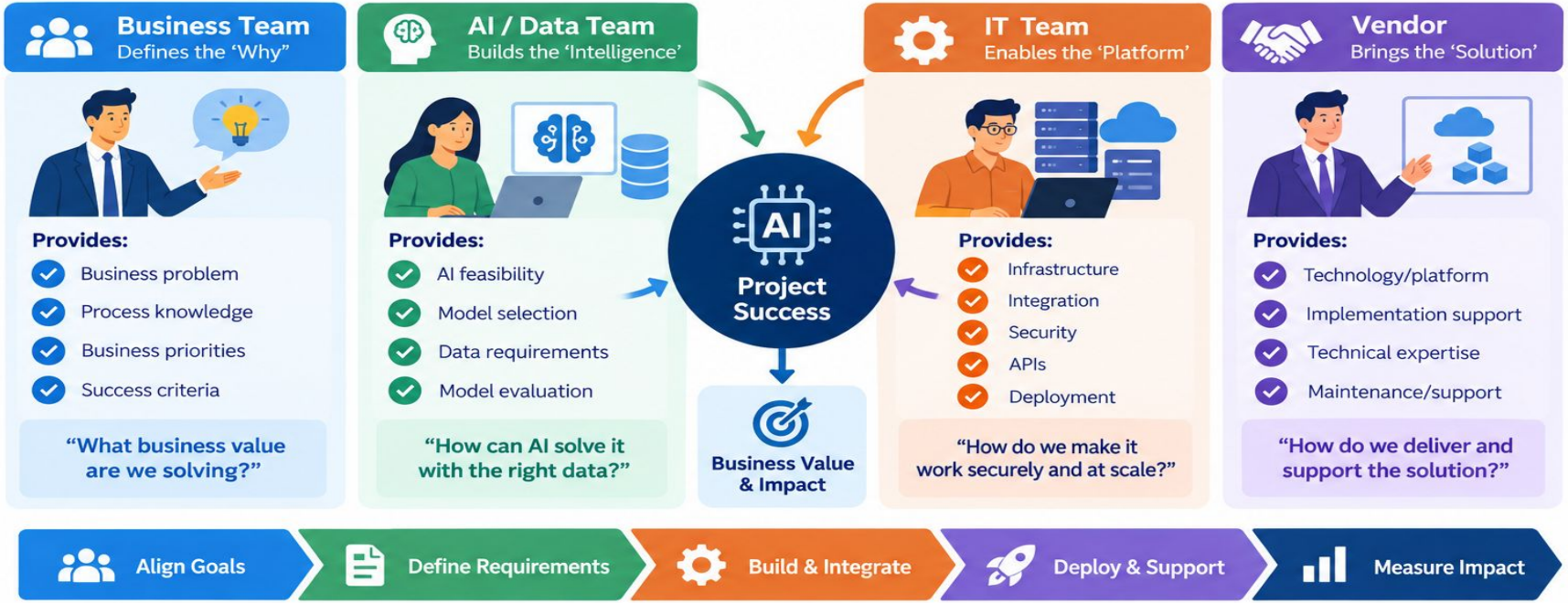
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Working With AI, IT & Vendors

Cross-Functional Collaboration : AI projects require collaboration between multiple teams.

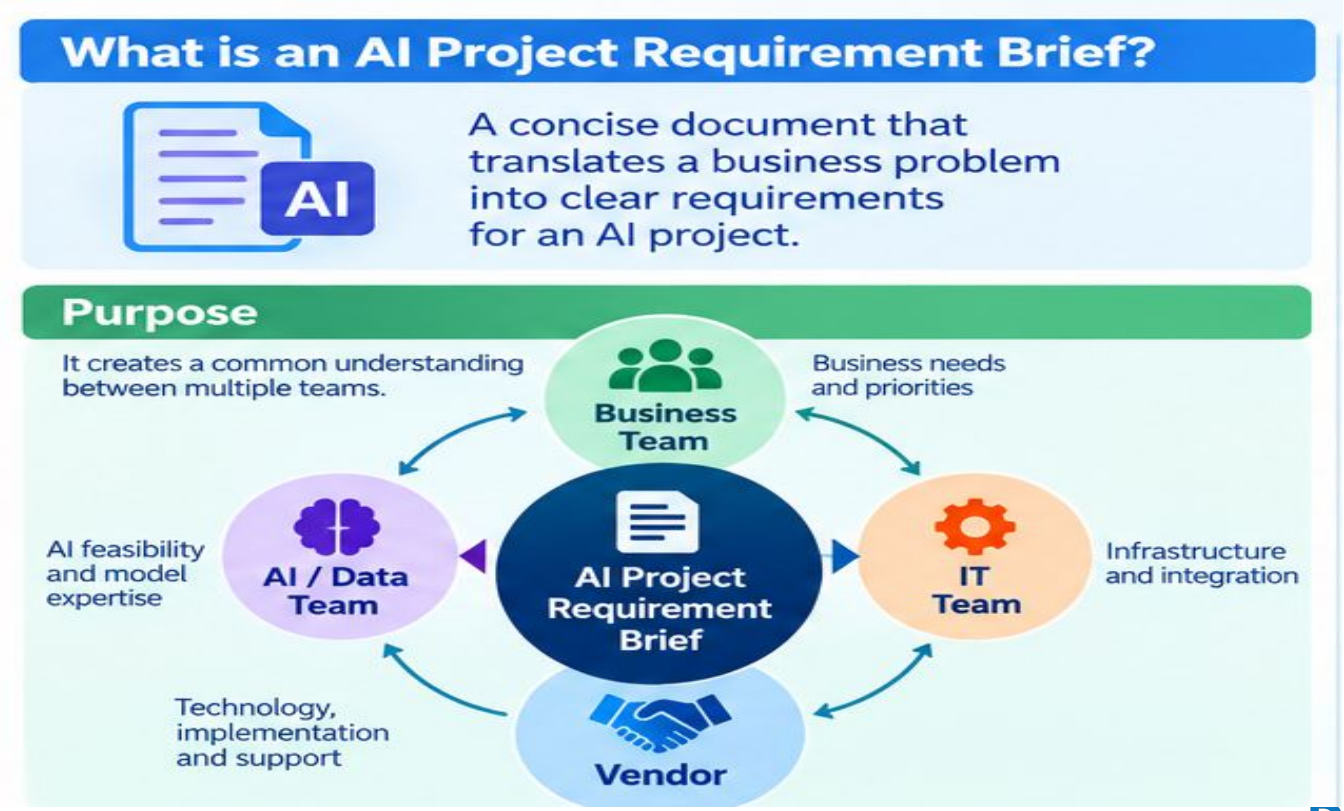


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AI Project Requirement Brief



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AI Project Requirement Brief - Template



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THANK YOU

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